

Health risk assessment of fluoride contamination in groundwater: A case study from the Mogamureru River basin, YSR Kadapa District, Andhra Pradesh, India

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ABSTRACT

Fluoride is essential for dental health in trace amounts; however, excessive exposure poses significant health risks. This study aims to evaluate the concentration and associated health risks of fluoride in groundwater in the Mogamureru River Basin, YSR Kadapa District, Andhra Pradesh, India. A total of 46 groundwater samples were collected from bore wells during the pre-monsoon (PRM) and post-monsoon (POM) seasons. Physicochemical parameters were analyzed, and multivariate statistical techniques, including Principal Component Analysis (PCA), were employed to identify the major contributors to groundwater contamination. Results revealed elevated levels of fluoride and nitrate, attributed to geogenic sources and agricultural activities. Health risk assessments were conducted using the United States Environmental Protection Agency (USEPA, 2014) methodology, by calculating the Hazard Quotient (HQ) and Total Hazard Index (THI) for adults, children, and infants. Fluoride concentrations exceeded the BIS permissible limit (1.5 mg/L) in 93% (PRM) and 95% (POM) of samples. Moreover, HQ values exceeded unity in 97% of adult, 98% of child, and 98% of infant samples, indicating significant non-carcinogenic health risks. Water Quality Index (WQI) analysis classified 58.7% (PRM) and 67.4% (POM) of samples as unsuitable for drinking. Pollution Index of Groundwater (PIG) indicated that 82.6% of samples fall within moderate pollution zones in both seasons. The findings underscore the urgent need for mitigation strategies and groundwater quality monitoring to safeguard public health and water security in the region.

Keywords: Water quality index; Pollution Index of groundwater; Total hazard index; Hazard quotient, Fluoride contamination, Mogamureru River Basin

INTRODUCTION

Access to clean and safe drinking water is fundamental to human survival and well-being. However, the growing demand for potable water, driven by population growth and urbanization, continues to outpace supply, leading to increasing challenges in maintaining a sustainable balance. Compounding this issue is the escalating contamination of water resources, which significantly reduces the availability of clean drinking water (Fida et al., 2023; Mishra, 2023). Both natural processes and anthropogenic activities contribute to the degradation of water quality. While certain chemical constituents in groundwater are essential for human health, their presence above permissible limits can pose serious health hazards. In recent years, environmental quality, particularly in relation to human health, has emerged as a major global concern (Malik et al., 2024; Reddy et al., 2024; Lodh et al., 2025). This growing awareness underscores the urgent need for systematic monitoring and assessment of groundwater quality to ensure the protection of public health and environmental sustainability.

Fluoride is commonly found in various environmental media, including water, air, soil, rocks, dust, food, and personal care products such as toothpaste and mouthwash (Raghav et al., 2024). Its widespread use in insecticides and rodenticides, has led to long-term environmental persistence. While fluoride plays a beneficial role in preventing dental problems, increasing concerns have emerged regarding its overexposure, particularly through contaminated groundwater and the excessive fluoridation of drinking water supplies. In both surface and groundwater systems, fluoride is typically introduced through the natural leaching of fluoride-bearing rocks and soils as water flows through geological formations (Badapalli et al., 2024). Additionally, anthropogenic sources such as industrial effluents and improper domestic waste disposal exacerbate fluoride contamination in groundwater.

Globally, fluoride pollution in drinking water has become a critical environmental and public health concern. Consequently, several countries have reevaluated and, in many cases, discontinued aggressive water fluoridation programs due to the associated health risks. The United States Environmental Protection Agency (USEPA, 2014) has developed a standardized approach known as human health risk assessment to evaluate the adverse effects of fluoride exposure, even at relatively low concentrations (e.g., 1.2 mg/L), on populations residing in contaminated areas.

The primary objective of this study is to: (i) analyze the physicochemical characteristics of tube well water, (ii) assess the spatial distribution and potential health risks associated with fluoride

exposure among the local population, (iii) evaluate the suitability of groundwater for drinking purposes using the Water Quality Index (WQI), and (iv) apply the Pollution Index of Groundwater (PIG) to determine the degree of groundwater contamination.

Geologically, the study area is predominantly characterized by quartzite–carbonate–shale sequences. The region is also known for dormant Barytes and yellow ochre mines, along with several road metal quarries. It falls within the Papaghni and Chitravathi groups of the Lower Cuddapah Supergroup (Golla et al., 2021; Sunitha et al., 2022). The lithological units include basic flows and igneous intrusions, accompanied by dolerite, quartzite, dolomite, quartzite with conglomerates, and chert along bedding planes. Pulivendula quartzites occur as narrow bands separating the Vempalle limestone and shale from the Tadipatri shale. Geomorphologically, the area exhibits denudational hills, pediments, pediplains and structural hills. Groundwater is predominantly stored within weathered zones and the joints and fractures of the Papaghni and Chitravathi rock groups. Massive limestone and Gulcheru quartzite formations serve as potential aquifers.

The region is heavily influenced by textile industries, and the quality of groundwater has significantly deteriorated due to urbanization, industrialization, and intensive agricultural and mining activities. In urban centres, textile effluents and domestic sewage are the primary sources of groundwater contamination. Therefore, this study aimed to assess the fluoride concentrations in groundwater samples collected from 46 settlements across the Mogamureru River Basin.

In addition to chemical analysis, fluoride levels were evaluated for their potential health implications. The results indicated that elevated fluoride concentrations are attributed to both geogenic and anthropogenic sources. Health risk assessment revealed non-carcinogenic risks for all age groups, infants, children, and adults. The findings offer critical insights for future water de-fluoridation efforts and public health protection strategies.

STUDY AREA

The Mogamureru River Basin, covering an area of 408.6 km², is located in the southwestern part of YSR Kadapa District, Andhra Pradesh, India. The basin lies between 14°10' to 14°40' N and 78°10' to 78°30' E, as shown in Survey of India Toposheet Nos. 37 J/2, 37 J/3, 37 J/6, 37 J/7, 37 J/11, 37 F/14, and 37 F/15 (Figure 1). Climatically, the region experiences arid to semi-arid conditions, with annual rainfall ranging between 600 and 650 mm, primarily influenced by the southwest and northeast monsoons. Temperature varies seasonally, ranging from 20.4°C during

winter to a peak of 43°C in summer. The depth to the water table in the region varies between 10 and 30 m, depending on the topography and seasonal recharge conditions.

Lithologically, the area is dominated by quartzite, dolomite, and shale formations, which significantly influence the hydrogeological characteristics of the region. The study area geologically falls within the Papaghni and Chitravathi Groups of the Lower Cuddapah Supergroup (Rao, 2025). The general strike of the rocks varies across the basin: trending north-eastward in the southern part, eastward in the central region, and southward in the northern sector. The dominant geomorphic features include denudational hills, structural hills, pediments, and pediplains, which control surface runoff and infiltration. The region supports mixed agricultural practices, with major crops including paddy, groundnut, sunflower, and sweet lime. The interplay of geological formations, climatic variability, and intensive land use has a direct impact on the groundwater quality and availability within the basin.

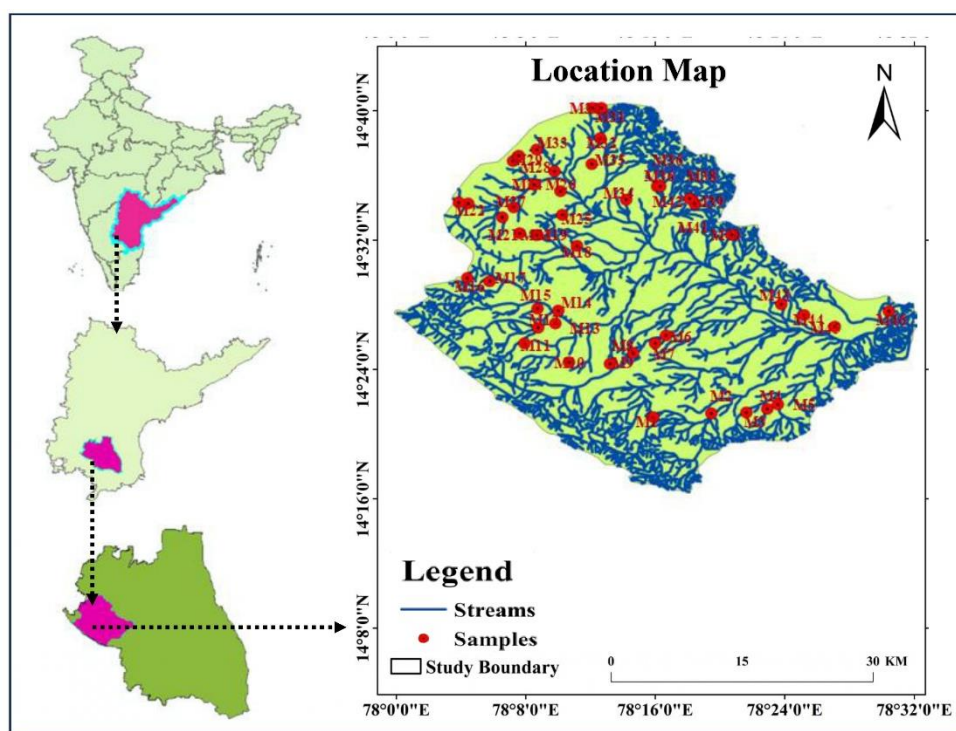


Figure 1. Sample site map of the investigated Mogamureru River Basin, located in the southwestern part of YSR Kadapa District, Andhra Pradesh, India

METHODOLOGY

Water samples and Analytical procedures

Groundwater samples were collected from 46 operational tube wells across the study region. Sampling was conducted randomly during the pre- and post-monsoon seasons of 2023 to capture seasonal variations in water quality. The sampling locations were strategically distributed across the Mogamureru River Basin, ensuring adequate spatial representation and accounting for proximity within the water distribution system.

Each sampling site was georeferenced using a Global Positioning System (GPS) to accurately record its coordinates. Samples were collected in 250 mL high-density polypropylene (HDPE) bottles. Prior to collection, the bottles were thoroughly cleaned with distilled water and rinsed multiple times with groundwater from the respective site to minimize contamination and ensure sample integrity (Gugulothu et al., 2022; Golla and Badapalli, 2022). The collected samples were then subjected to physicochemical analysis, and the data obtained were used to assess human health risks associated with fluoride exposure. The methodological framework adopted for health risk assessment is illustrated in Figure 2.

Groundwater quality parameters/constituents were analyzed according to conventional procedures (Clesceri, 1998), and actions were proposed to avoid contamination. During groundwater sampling, a portable pH/EC/TDS meter (Hanna HI 9811-5) was used to detect pH, electrical conductivity (EC), and total dissolved solids (TDS). Titration was used in the laboratory to quantify total hardness (TH), bicarbonate (HCO_3^-), carbonate (CO_3^{2-}), chloride (Cl^-), calcium (Ca^{2+}), and magnesium (Mg^{2+}). Sulphate (SO_4^{2-}) and nitrate (NO_3^-) concentrations were measured using a UV-visible spectrophotometer (Spectronic, 21, BAUSCH and LOMB). The sodium (Na^+) and potassium (K^+) concentrations were measured using a flame photometer (Systronics, 130). The fluoride (F^-) was measured using the "Mohr's technique and Fluoride Analyzer" ion-selective electrode (ISE).

Analytical accuracy

Groundwater samples were obtained in during two different seasons: pre-monsoon (PRM) and post-monsoon (POM) seasons. We used standard protocols to evaluate the collected samples for major and minor dissolved ions. The total cation and total anion balances were utilized to calculate the ionic charge balance error % (Freeze and Cherry, 1979). The sample error percentage is around $\pm 5\%$, with some outliers due to seasonal variations in ion levels. Groundwater chemical analysis errors can be caused by a variety of variables, including the use of reagents, limitations in

the procedures and instruments used, the presence of contaminants in distilled water, and other possible sources.

$$IBE = \frac{\sum Cations - \sum Anions}{\sum Cations + \sum Anions} \times 100(1)$$

Water quality index (WQI)

The WQI is frequently used to assess the quality of groundwater and its suitability for various applications, including ingestion and irrigation. Numerous researches have used the WQI approach to evaluate the quality of groundwater for drinking and irrigation in various parts of the world (Horton, 1965; Brown et al., 1972; Latif et al., 2024; Nayak et al., 2024). The relevant (WQI) parameters and formulas are presented in Table 1. The current study used equations to determine 12 parameters for WQI: pH, EC, TDS, TH, Cl⁻, Na⁺, HCO₃⁻, Ca²⁺, Mg²⁺, F⁻, NO₃⁻, and SO₄²⁻.

Health Risk Assessment

Consuming contaminated drinking water poses several health risks. Therefore, assessing the potential non-carcinogenic health effects associated with fluoride exposure is essential. This study follows the U.S. Environmental Protection Agency (USEPA, 2014) health risk assessment protocol, as applied in recent studies (Alam et al., 2024). Given fluoride's dual role as a vital nutrient at low concentrations and a potential toxin at elevated levels, it was selected as the key contaminant for health risk evaluation. The risk assessment focused on three population groups: infants, children, and adults, within the Mogamureru River watershed. Exposure parameters were based on daily water intake specific to each age group.

The health risk assessment involved calculating the Average Daily Dose (ADD) and HQ of fluoride through ingestion and dermal absorption (Table 2). The key variables include; C: Fluoride concentration in groundwater (mg/L); SA: Skin surface area (cm²); Kp: Skin permeability coefficient (cm/h); EF: Exposure frequency (days/year) - assumed as 365 days for all age groups; ED: Exposure duration (years); ET: Exposure time (h/day); AT: Averaging time (days); IR: Water ingestion rate (L/day) (Adimalla and Qian, 2023; Abdipour et al., 2025)

Table 1.WQI, Human health risk and pollution parameters with formulae

Parameter	Formulae
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Water Quality Index (WQI)	$WQI = \sum_{i=1}^n Sli$ Sli is the sub index of each parameter $\sum_{i=1}^n Sli$ is the sum of the Sli
	$Sli = W_i \times q_i$ W_i is relative weights q_i is a rating of quality
	$q_i = \left(\frac{C_i}{S_i} \right) \times 100$ C_i is the concentration of each parameter S_i is the BIS recommended value for each parameter
	$WQI = \frac{W_i q_i}{W_i}$ W_i is relative weights q_i is a rating of quality
Average daily dose for Dermal	$ADD_{dermaal} = C \cdot K_p \cdot S_A \cdot \frac{ED \cdot EF \cdot ET \cdot CF}{BW \cdot AT}$ C stands for fluoride/nitrate pollution in groundwater (mg/L) K_p stands for sun permeability coefficient cm/h S_A stands surface area cm^2 ED stands for exposure duration years EF stands for exposure frequency ET stands for an exposure time l/cm³ AT means average exposure duration in days
Average daily dose for ingestion	$ADD_{ingestion} = \frac{C \cdot IR \cdot ED \cdot EF}{BW \cdot AT}$ C stands for fluoride/nitrate pollution in groundwater (mg/L) IR stands for ingestion rate ED stands for exposure duration years EF stands for exposure frequency AT means average exposure duration in days
Hazard quotient (HQ)	$HQ = \frac{ADD}{Rfd}$ ADD is average daily dose dermal/ ingestion Rfd is the oral reference dose mg/kg/day
Hazard index (HI)	$HI = \sum_{i=1}^n HQ$ HQ is the Hazard quotient $\sum_{i=1}^n HQ$ is the sum of Hazard quotient
Pollution index (PIG)	$PIG = \sum O_w$ $\sum$ is the sum of overall water quality O_w is for overall water quality
	$O_w = W_p \times S_c$ Where W_p represents for weight ingredient, S_c stands for concentration condition

	$S_c = \frac{C}{D_s}$ C means the chemical variable concentration of a specific sample Ds means the standard
	$W_p = \frac{R_w}{\sum R_w}$ Rw is the weight of each parameter $\sum R_w$ is the sum of the weight of each ingredient

The HQ is computed as the ratio of ADD to the Reference Dose (RfD). According to the USEPA, a Hazard Index (HI) > 1 indicates that the non-carcinogenic risk exceeds the acceptable limit, implying potential adverse health effects. Conversely, an HI < 1 signifies that the fluoride concentration in drinking water is within the safe range for human health across all age groups. The findings from this assessment aim to inform public health protection strategies and support effective defluorination planning.

Table 2. Health Risk Assessment parameters

Parameters(units)	Infants	Children	Adults	References
Average Daily Dose Dermal				
Sun Permeability Coefficient (KP) cm ² /h	0.001			Adimalla and Qian (2023)
Surface Area (SA) cm ²	4500	10800	19800	
Exposure Duration (ED) Y	1	11	65	
Exposure Frequency (EF)D/Y	365			
Exposure Time (ET)	0.6 h/d			
Conversation Factor (CF)	0.001 l/cm ³			
Average Exposure Duration (AT)D	365	4015	23725	
Average Daily Dose Ingestion				
Ingestion Rate (IR) l/d	0.36	0.414	1.046	Abdipour et al. (2025)
Exposure Duration (ED) Y	1	11	65	
Exposure Frequency (EF)D/Y	365			
Average Exposure Duration. (AT)D	365	4015	23725	

The Pollution Index of groundwater (PIG)

The pollution index considers all water quality indicators, including pH, EC, TDS, TH, Cl^- , HCO_3^- , Na^+ , Ca^{2+} , Mg^{2+} , NO_3^- , F^- , and SO_4^{2-} , in relation to one another. The groundwater contamination index was established using multiple procedures (PIG) (Sanad et al., 2024). The PIG is calculated using the formula in Table 1.

Statistical Analysis

Principal Component Analysis (PCA)

PCA is a statistical technique used to reduce data dimensionality while retaining as much information as possible. It captures the majority of variability among data points and is widely recognized as an effective method for simplifying complex datasets. PCA preserves most of the essential information while minimizing complexity (William and Katambara, 2025). In this study, pollution sources were identified using PCA, which was performed with XL-Start software.

Correlation Analysis

Correlation analysis is a statistical method used to evaluate both the strength and direction of relationships between two variables. In hydrochemical studies, positive correlation coefficients indicate that certain geogenic or anthropogenic processes strongly influence water quality parameters, whereas negative correlations suggest limited or inverse interactions. In this study, Pearson's correlation coefficient (r) was applied to examine the relationships among various physicochemical parameters of groundwater. The analysis was performed at a 95% confidence level ($\alpha = 0.05$) to identify statistically significant correlations. This approach helps in tracing potential sources of contamination and understanding interactions among different water quality variables.

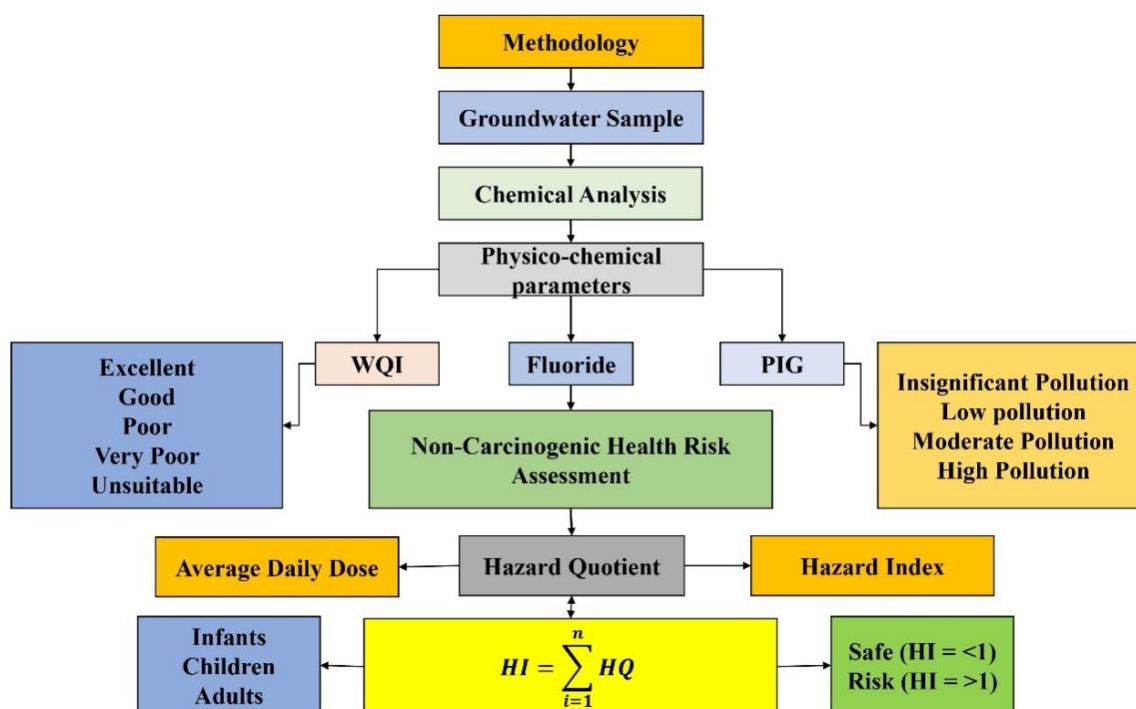


Figure 2. Flowchart indicating methodological frame work used in the present Investigation.

RESULTS AND DISCUSSIONS

Groundwater exhibits an alkaline pH of 7.5 to 8.6 pre-monsoon and 7.2 to 8.5 post-monsoon, with 94% and 90% of samples conforming to WHO (2012) and BIS (2012) standards, respectively. Electrical conductivity (EC) ranged from 720 to 4000 $\mu\text{S}/\text{cm}$ pre-monsoon and 632 to 6820 $\mu\text{S}/\text{cm}$ post-monsoon, with 65% and 67% exceeding WHO (2012) guidelines. Most samples were classified as brackish water, with total dissolved solids (TDS) from 1350.6 to 2295.8 mg/L pre-monsoon and 764.7 to 2152.6 mg/L post-monsoon (Table 3). Seasonal variations in EC and TDS are attributed to rain dilution. Post-monsoon, total hardness increases due to higher calcium and magnesium ion levels. High bicarbonate levels in several samples suggest sulfate reduction reactions. Elevated calcium correlates with low rainfall and poor aquifer recharge, while magnesium varies seasonally. Potassium concentrations exceed standards in both seasons, notably higher in dry periods, raising concerns over potential health risks such as dental and skeletal issues, thyroid disorders, neurological problems, acne, and cardiovascular complications.

Table 3. Physicochemical parameters (WHO, 2012; BIS, 2012).

Parameter	WHO (2012)	BIS (2012)	PRM samples	POM samples	WHO (2012)		BIS (2012)	
					PRM samples (%)	POM samples (%)	PRM samples (%)	POM samples (%)
pH	7.0–8.5	6.5–8.5	7.5-8.6	7.2-8.5	94	90	94	90
EC	750-1500	-	720-4000	632-6820	65	67	-	-
TDS	500-1500	500-2000	1350.6-2295.8	764.7-2152.6	73	89	95	69
TH	100-500	300-600	0-168	465.0-1421.3	13	95	2	91
HCO ₃ ⁻	500	-	428.4-762	268.4-695.4	57	97	-	-
Cl ⁻	200-600	250-1000	162.9-429.9	89.9-569.8	100	100	100	100
Ca ²⁺	75-200	75-200	138.65-398.7	72.1-376.7	15	10	15	10
Mg ²⁺	50-150	30-100	63.5-161.2	38.5-131.22	84	30	80	50
K ⁺	12	-	12.2-72.9	1.22-19.71	100	56	-	-
SO ₄ ²⁻	200-400	200-400	145.2-381.8	57.2-351.8	50	70	50	70
Na ⁺	200	-	48.4-236.4	1.22-196.4	50	40	-	.
F ⁻	1.5	1-1.5	1.55-5.01	1.05-4.85	97	97	100	100
NO ₃ ⁻	45	45	6.7-70.7	1.60-11.3	60	70	60	70

Fluoride and Nitrate hotspots in the study region

The dominant nitrogenous compound in the groundwater of the study area is nitrate (NO₃⁻), with approximately 60% of pre-monsoon and 70% of post-monsoon (Table 4) samples exceeding WHO (2012) and BIS (2012) permissible limits. The variability in NO₃⁻ concentrations indicates diverse nitrogen sources primarily from anthropogenic activities such as fertilizer and sewage seepage. Furthermore, fluoride (F⁻) levels in 97% of samples surpassed the WHO threshold of 1.5 mg/L (WHO, 2012), indicating a more severe health threat than nitrate. Fluoride contamination derives from both natural (fluorite dissolution) and anthropogenic sources like sewage infiltration and chemical fertilizers.

Water Quality Index

The Water Quality Index (WQI) was utilized to categorize groundwater quality, revealing values from 49.0 to 1522.1 pre-monsoon and 20.0 to 1546.0 post-monsoon. Classification results showed that 2.2% (pre-monsoon) and 4.4% (post-monsoon) of samples were classified as “excellent,” while 6.5% and 2.2% were “good,” 8.7% and 13.0% “poor,” 23.9% and 13.0% “very poor,” and 58.7% and 67.4% “unsuitable” (Table 5). The increase in unsuitable groundwater post-monsoon is noted, possibly due to ion leaching, groundwater overexploitation, effluent discharge, and agricultural practices. Several villages showed high WQI concentrations, with spatial distribution illustrated in Figure 3.

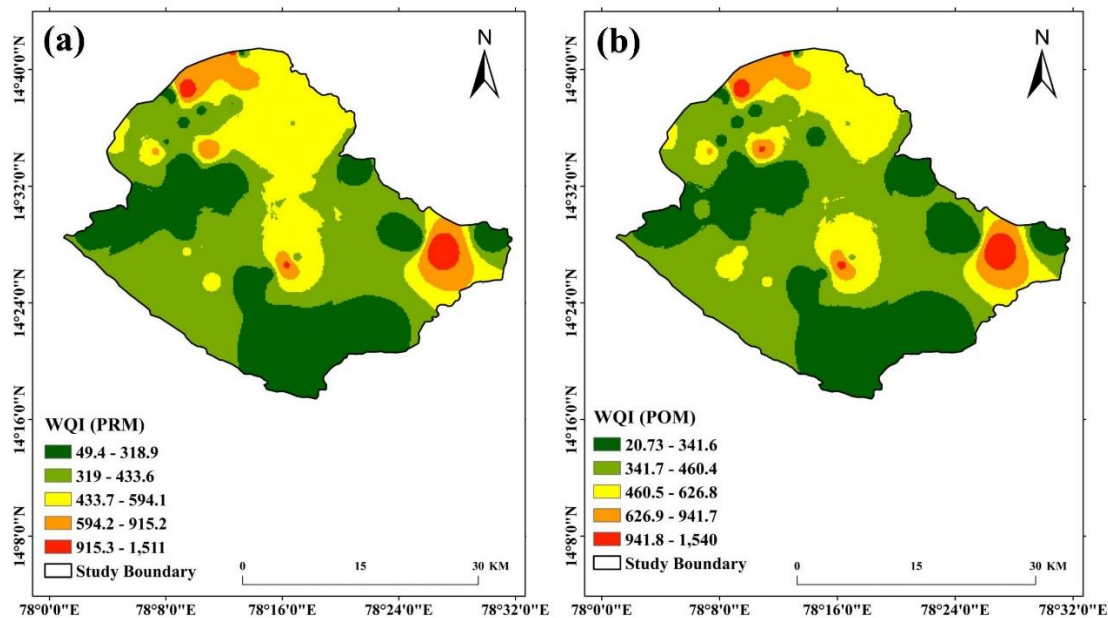


Figure 3. Spatial distribution map of WQI in the study area

(a) Pre-monsoon (b) Post-monsoon

Pollution index of groundwater

The PIG was utilized to evaluate groundwater and surface water quality, revealing that 8.7% of pre-monsoon and 6.5% of post-monsoon samples fell into the insignificant contamination zone (<1). Additionally, 4.3% of pre-monsoon and 6.5% of post-monsoon samples were classified in the low contamination zone (1-2). The majority, 82.6% across both seasons, were categorized within the moderate contamination zone (2-2.5), while 2.17% of samples in each season were in the high contamination zone (>2.5) (Table 5). Spatial distributions of PIG values for both seasons are depicted in Figure 4.

Table 4. Fluoride concentration in water samples in various villages

Villages	Sources	Coordinates		F ⁻ concentration (mg/L)	
		X	Y	PRM samples	POM samples
Simhadripuram	Well	13.98739	78.697	1.55	1.05
Jangamreddypalle	Well	14.22706	78.69178	5.01	4.85
Teliki	Well	14.22564	78.74164	3.42	3.20
Sunkesula	Well	14.06503	78.74308	4.40	4.00
Diddekunta	Well	14.13147	78.81203	1.80	1.60
Kasanur	Well	14.11469	78.86525	2.02	1.80
Gurjala	Well	14.06	78.755	2.80	2.60
Bidenamcherla	Well	14.0855	78.91486	2.70	2.30
Himakuntla	Well	14.09344	78.85639	3.10	2.90
AnkalammaGudur	Well	13.95503	78.78389	2.30	1.95
Lomada	Well	14.06239	78.77736	3.60	3.15
Balapanur	Well	13.89883	78.66522	2.85	2.65
Agraharam	Well	14.05089	78.64256	4.90	4.60
Akkulagaripalle	Well	14.03922	78.7405	3.12	2.85
Bonala	Well	13.99108	78.63111	2.60	2.60
Ippatla	Well	14.05	78.80783	2.65	2.45
Lingala	Well	13.94167	78.68346	3.65	3.35
PeddaKudala	Well	14.02	78.64425	3.92	3.70
ChinnaKudala	Well	13.98633	78.76772	4.90	4.70
Kamasamudram	Well	14.12036	78.95114	3.95	3.85
Gunkanapalle	Well	13.88681	78.62094	3.10	2.90
Loputnuthula	Well	13.9	78.68919	3.05	2.85
Theranampalle	Well	14.07461	78.91786	4.25	3.95
Ankenipalli	Well	14.00931	78.58528	3.85	3.65
Cherlopalle	Well	14.22611	78.73872	3.95	3.75
Inaganur	Well	14.227	78.62961	3.45	3.05
Mallela	Well	13.96378	78.84636	3.90	3.70
Bhadrampalle	Well	14.20258	78.64267	4.20	3.85
Bhuchupalle	Well	13.94014	78.74453	4.10	3.95
Thondur	Well	14.19953	78.91467	3.70	3.40
Nallecherupalle	Well	13.99331	78.83831	2.95	2.80
Velpula	Well	14.02578	78.86028	3.90	3.60
V.Kottapalle	Well	14.106	78.77825	4.25	4.05

Vemula	Well	14.15769	78.85808	4.35	4.20
Vempalli	Well	13.96625	78.83067	4.45	4.25
Ramireddypalle	Well	14.23069	78.95678	4.45	4.20
Thallapalle	Well	13.93208	78.65042	4.35	4.15
Yerraballe	Well	14.26389	78.75247	4.35	4.15
Ulimella	Well	14.14578	78.68333	4.23	4.00
Pulivendula	Well	14.2725	78.72322	3.52	3.30
Gotur	Well	14.16519	78.77806	4.25	4.10
Venkatapuram	Well	14.21075	78.76972	4.10	3.90
Palagiri	Well	14.08661	78.67719	4.70	4.50
Veerapunayanipalle	Well	14.10453	78.62858	4.25	4.00
Moillacheru	Well	14.17825	78.72825	4.10	3.90
Animela	Well	14.14458	78.73125	4.35	4.10

Table 5. Classification of water quality types based on WQI (Brown et al., 1972)
And PIG (Subba Rao, 2012)

WQI				PIG			
Range	Water quality status	% of water samples (PRM)	% of water samples (POM)	Classification	class	PRM Samples (%)	POM Samples (%)
< 50	Excellent	2.2	4.4	<1	Insignificant pollution	8.7	6.5
50.1-100	Good	6.5	2.2	1- 2	Low pollution	4.3	6.5
100.1-200	Poor	8.7	13.0	2-2.5	Moderate pollution	82.6	82.6
200.1-300	very poor	23.9	13.0	>2.5	High pollution	2.17	2.17
>300	Unsuitable for drinking	58.7	67.4	-	-	-	-

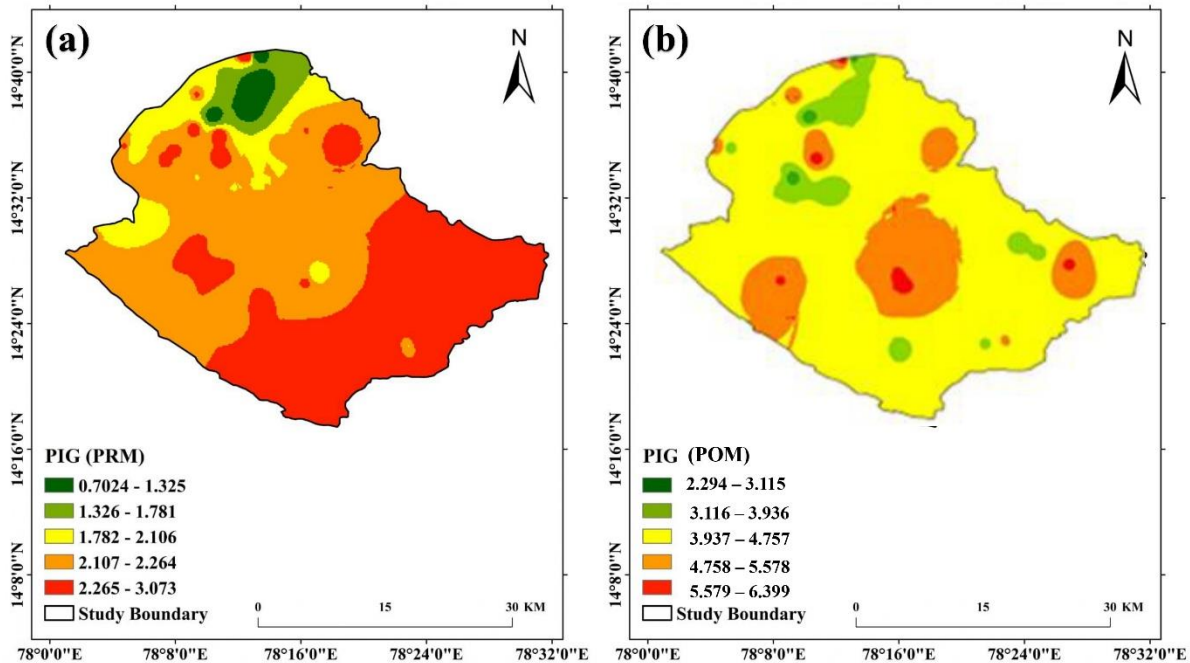


Figure 4. Spatial distribution map of PIG in the study area
(a) Pre-monsoon and (b) Post-monsoon

Health risk assessments for dermal and drinking

Given that several rural water supply wells in the study area exhibit elevated fluoride levels, assessing fluoride exposure and associated health risks (FEHRA) is crucial (Adams and Garman, 2024; Abdipour et al., 2025). This study evaluated the effects of fluoride on human health using the HQ, a widely used metric for quantifying non-carcinogenic risks in humans. Excessive fluoride intake can lead to various health issues (Figure 5).

The HQ and HRA data indicate that THI cutaneous samples from infants, children, and adults consistently fall within safe zones before and after the monsoon. Figure 6, suggesting minimal dermal risk from THI exposure (Table 6 and 7). However, ingestion samples reveal a significantly higher risk, especially pre-monsoon, where 98% of samples are hazardous. Post-monsoon ingestion samples show slight improvement but still predominantly hazardous, particularly for infants and children (Figure 7) (95.46% hazardous). These results highlight fluoride ingestion as a critical health risk across all age groups (Figure 8). In contrast, nitrate and fluoride concentrations do not significantly contribute to dermal exposure risk, as indicated by the HRA results (Figure 9). Building on these findings, the ingestion pathway was further examined for the most vulnerable group infants. Figure 10 illustrate the spatial distribution of ingestion-

based HRA for infants in the study area, revealing distinct seasonal variations in health risk levels during the pre-monsoon and post-monsoon periods. The results suggest that mitigation strategies should prioritize reducing fluoride ingestion particularly among vulnerable populations such as infants and children while dermal exposure poses comparatively lower concern.

Table 6. Non-carcinogenic risk of HQ through dermal contact and Ingestion (PRM and POM)

Purposes	People	Floride & Nitrate	HQ					
			PRM			POM		
			Min	Max	Mean	Min	Max	Mean
Dermal	Adults	HQN	0.001	0.009	0.00583	0	0.009	0.00226
		HQF	0.008	0.026	0.01893	0.005	0.025	0.01776
	Children	HQN	-0.015	-0.001	-0.0098	-0.015	-0.001	-0.0036
		HQF	-0.043	-0.013	-0.032	-0.042	-0.009	-0.0299
	Infants	HQN	0.001	0.007	0.00474	0	0.007	0.00193
		HQF	0.006	0.02	0.0147	0.004	0.019	0.01376
Ingestion	Adults	HQN	0.076	0.804	0.51478	0.03	0.804	0.19817
		HQF	0.705	2.278	1.6742	0.478	2.206	1.56604
	Children	HQN	0.093	0.978	0.6265	0.036	0.978	0.24107
		HQF	0.858	2.773	2.03752	0.581	2.684	1.90589
	Infants	HQN	0.089	0.941	0.60278	0.035	0.941	0.23209
		HQF	0.825	2.668	1.96048	0.559	2.583	1.83378

Table 7. Non-carcinogenic risk of HRA through dermal contact and Ingestion (PRM and POM)

Purposes	People	HRA					
		PRM			POM		
		Min	Max	Mean	Min	Max	Mean
Dermal	Adults	0.009	0.032	0.02487	0.006	0.03	0.02013
	Children	-0.053	-0.015	-0.04172	-0.051	-0.01	-0.03359
	Infants	0.007	0.025	0.019239	0.005	0.023	0.015478
Ingestion	Adults	0.781	2.795	2.188978	0.528	2.668	1.764174
	Children	0.951	3.401	2.664087	0.642	3.248	2.147
	Infants	0.915	3.272	2.563304	0.618	3.125	2.065783

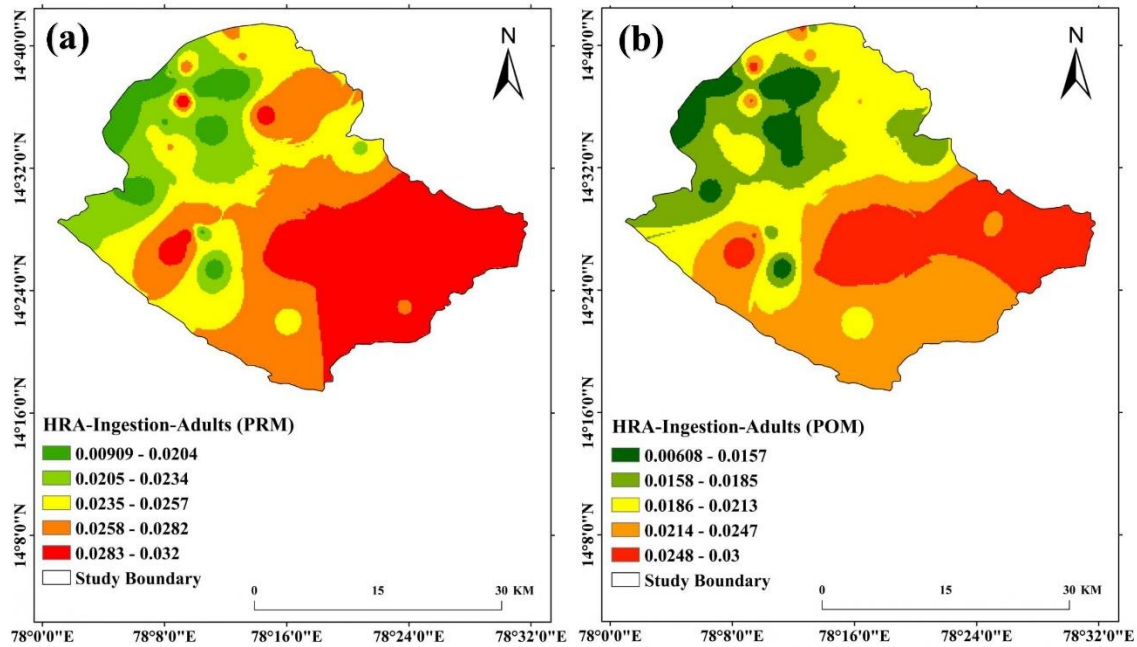


Figure 5. Spatial distribution map dermal of HRA for adults in the study area
(a) Pre-monsoon and (b) Post-monsoon

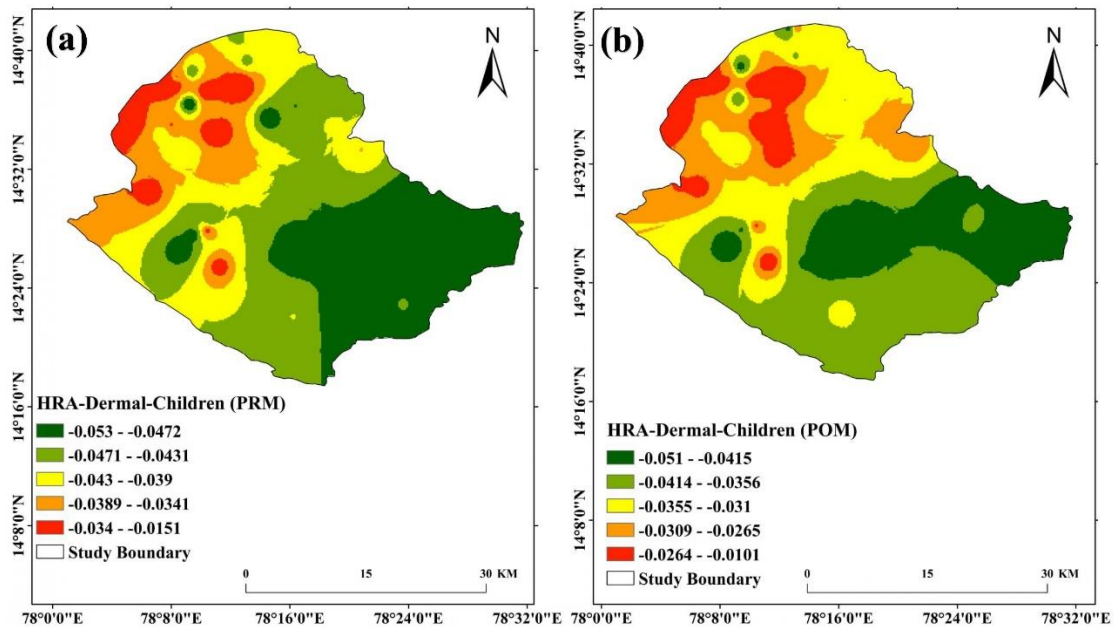


Figure 6. Spatial distribution map dermal of HRA for children the study area
(a) Pre-monsoon and (b) Post-monsoon

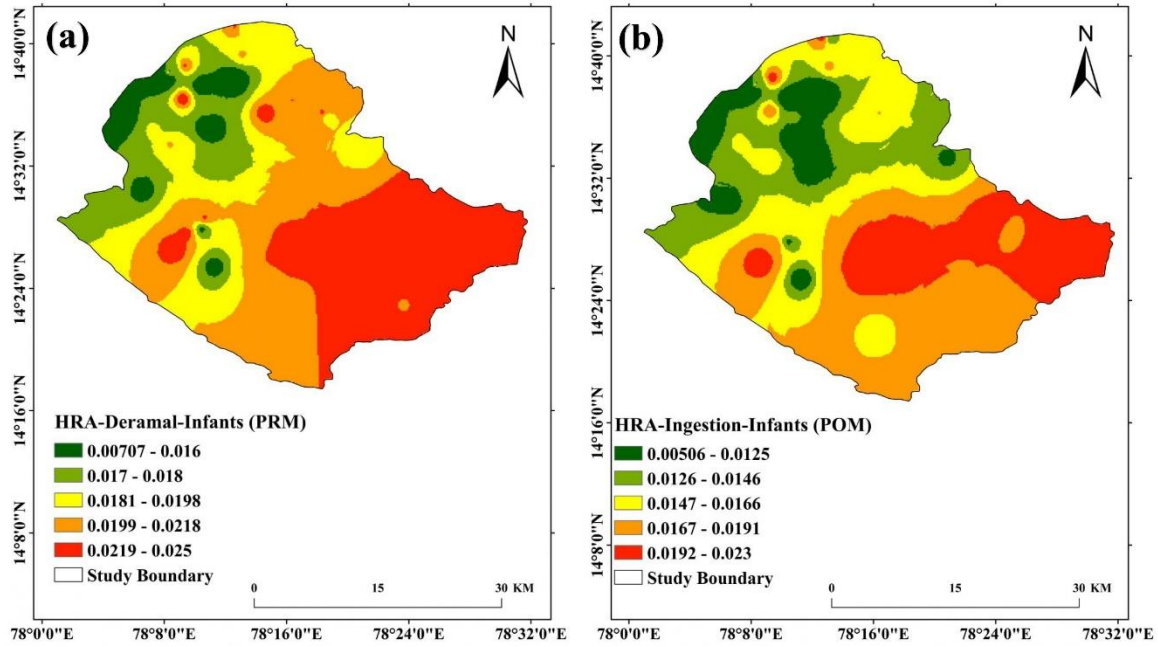


Figure 7. Spatial distribution map dermal of HRA for infants in the study area
(a) Pre-monsoon and (b) Post-monsoon

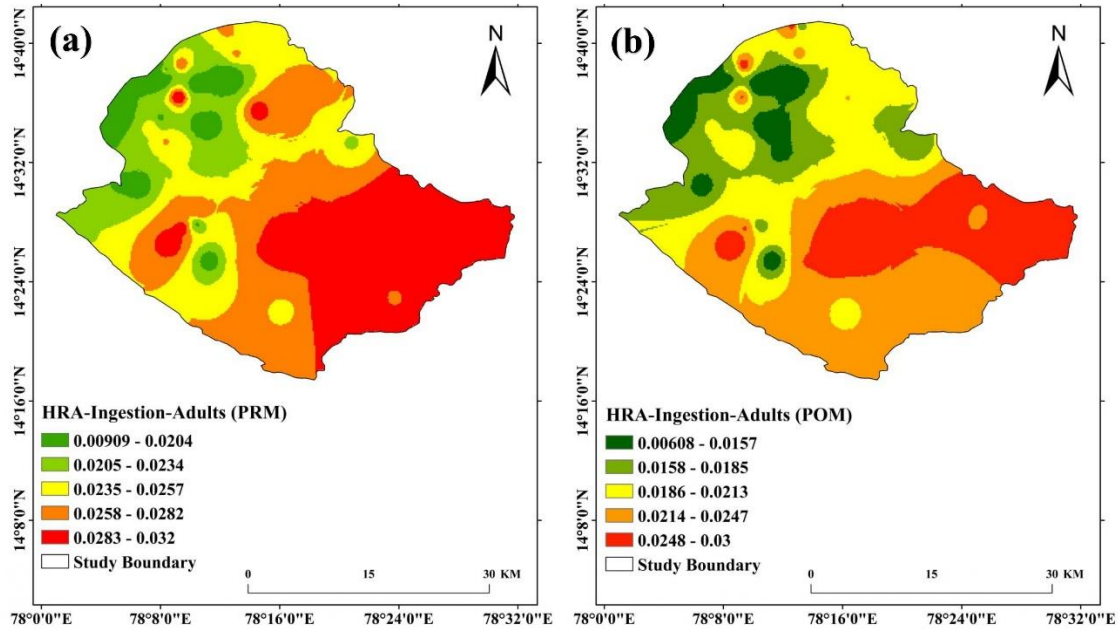


Figure 8. Spatial distribution map ingestion of HRA for adults in the study area
(a) Pre-monsoon and (b) Post-monsoon

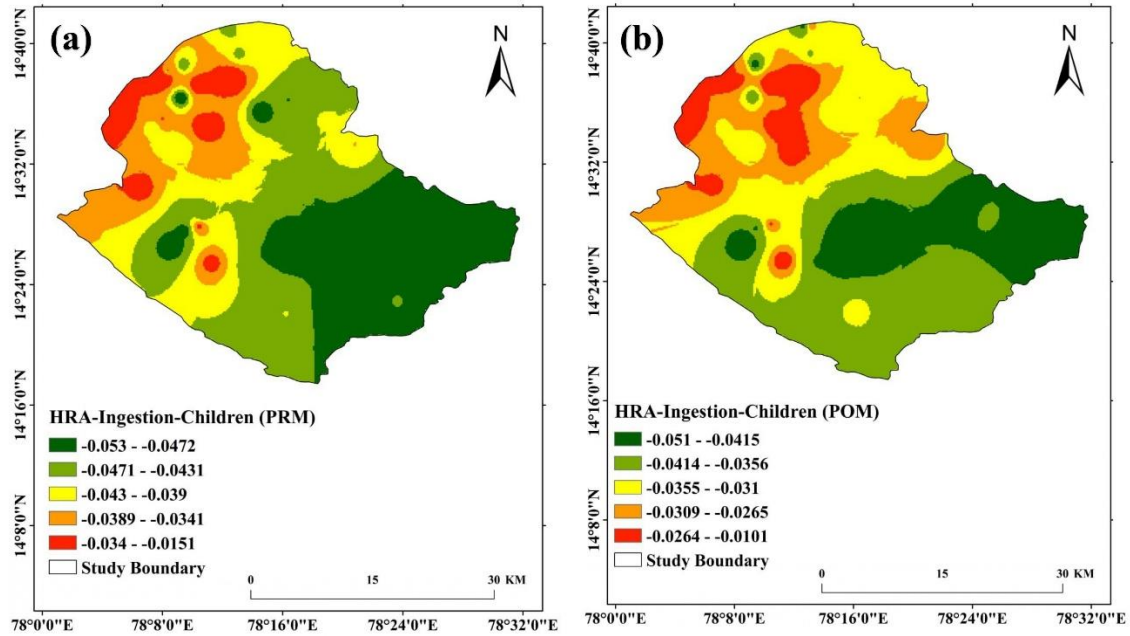


Figure 9. Spatial distribution map ingestion of HRA for children in the study area
(a) Pre-monsoon and (b) Post-monsoon

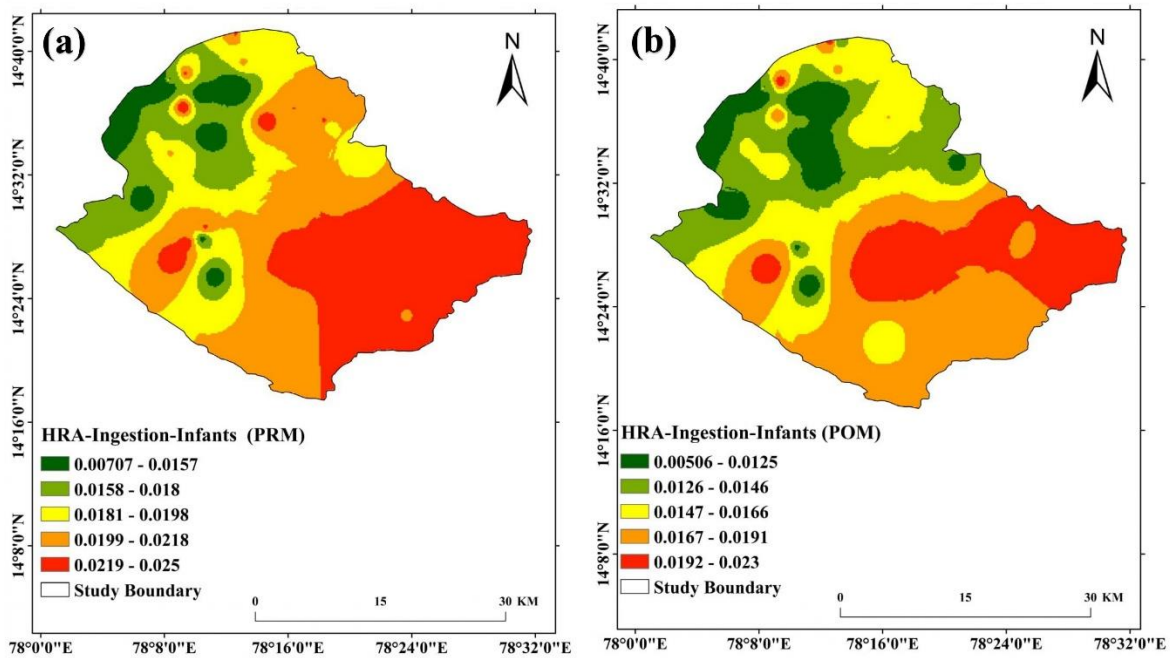


Figure 10. Spatial distribution map ingestion of HRA for infants in the study area
(a) Pre-monsoon and (b) Post-monsoon

Principal Component Analysis (PCA)

The PCA analysis of groundwater quality revealed that the first two components significantly represented the variability in water chemistry across study wells (Pandey et al., 2023). The pre-monsoon season identified eigen values of 6.168 and 2.472 for the first two components, explaining variances of 44.09% and 17.655% (Table 8) respectively, while during the post-monsoon, these values were 5.741 and 2.777 with variances of 41.005% and 19.834%. The first principal component (PC1) was heavily influenced by factors like EC, TDS, TH, Cl^- , Ca^{2+} , SO_4^{2-} , as well as Na^+ and NO_3^- , indicating effects of geological, hydrological, and anthropogenic sources. High concentrations of hardness and salinity were linked to local limestone and evaporite deposits, signalling natural mineral dissolution and the impact of surface runoff enriched by agricultural inputs. The second principal component (PC2) indicated geological influence and contamination from pit latrines. Subsequent components (PC3 to PC5) highlighted the contributions of agrochemicals, geological factors, and minor agricultural influences on groundwater variability. Overall, the analysis underscores the interplay between agricultural practices and natural geological processes affecting groundwater quality.

Table 8. PCA of selected parameters (PRM and POM)

Parameter	PRM					POM				
	F1	F2	F3	F4	F5	F1	F2	F3	F4	F5
pH	0.08	0.177	0.521	0.034	0.061	0.031	0.374	0.29	0.027	0.033
EC	0.918	0.003	0.014	0.007	0.004	0.954	0.008	0.007	0.011	0.003
TDS	0.956	0.001	0.008	0.007	0.004	0.959	0.008	0.006	0.013	0.003
TH	0.706	0.236	0.001	0	0	0.812	0.123	0.001	0.014	0.007
CO_3^{2-}	0.102	0.394	0.327	0.002	0.049	0.007	0.52	0.305	0.022	0.005
HCO_3^-	0.308	0.018	0.145	0.384	0.036	0.264	0.008	0.003	0.612	0.035
Cl^-	0.745	0.077	0	0.059	0.004	0.712	0.074	0.006	0.05	0.001
Ca^{2+}	0.65	0.069	0.014	0.009	0.102	0.789	0.023	0.033	0.003	0.006
Mg^{2+}	0.318	0.345	0.009	0.02	0.171	0.294	0.32	0.154	0.035	0.003
K^+	0.207	0.195	0.151	0.06	0.237	0.17	0.375	0	0	0.234
SO_4^{2-}	0.284	0.274	0.072	0.222	0.001	0.36	0.212	0.043	0.184	0.008
Na^+	0.578	0.192	0.003	0.009	0.024	0.343	0.45	0.006	0.063	0.014
F^-	0.021	0.34	0.146	0.191	0.356	0.002	0.217	0.158	0.067	0.424
NO_3^-	0.295	0.15	0.133	0.162	0.002	0.042	0.065	0.511	0.127	0.027
Eigenvalues	6.168	2.472	1.544	1.167	0.752	5.741	2.777	1.523	1.23	0.805
Variability (%)	44.059	17.655	11.028	8.335	5.373	41.005	19.834	10.876	8.782	5.752

Cumulative %	44.059	61.713	72.741	81.077	86.45	41.005	60.839	71.715	80.497	86.249
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Pearson's correlation

EC shows a strong positive correlation with TDS in both PRM ($r = 0.991$) and POM ($r = 0.993$) seasons, indicating a mutual dependency where increased TDS directly raises groundwater EC. Significant positive correlations exist between EC and ions such as SO_4^{2-} , Na^+ , Cl^- , CO_3^{2-} , and HCO_3^- , suggesting similar lithological sources due to water-rock interactions (Wagh et al., 2020). TH correlates positively with Mg^{2+} , HCO_3^- , Cl^- , Ca^{2+} , SO_4^{2-} , Na^+ , and F^- , with Mg^{2+} , Cl^- , and SO_4^{2-} being principal contributors to water hardness, especially post-monsoon. Calcium exhibits moderate to strong correlations with Mg^{2+} , Na^+ , and SO_4^{2-} , indicating their origin from mineral dissolution. A weak negative correlation with F^- suggests distinct geochemical behaviours. Furthermore, magnesium's correlations with NO_3^- and F^- (Tables 9 and 10), indicate potential anthropogenic impacts, while natural geogenic processes predominantly influence groundwater chemistry, particularly post-monsoon.

Table 9. Pearson correlation matrix of groundwater parameters for PRM season

	pH	EC	TDS	TH	CO_3^{2-}	HCO_3^-	Cl^-	Ca^{2+}	Mg^{2+}	K^+	SO_4^{2-}	Na^+	F^-	NO_3^-
pH	1													
EC	0.306	1.000												
TDS	0.315	0.991	1.000											
TH	0.057	0.782	0.805	1.000										
CO_3^{2-}	0.655	0.350	0.348	-0.042	1.000									
HCO_3^-	-0.043	0.599	0.610	0.502	-0.079	1.000								
Cl^-	0.302	0.815	0.827	0.585	0.421	0.256	1.000							
Ca^{2+}	0.178	0.733	0.767	0.874	-0.051	0.538	0.614	1.000						
Mg^{2+}	-0.121	0.548	0.547	0.781	-0.014	0.264	0.326	0.380	1.000					
K^+	-0.046	-0.302	-0.344	-0.223	-0.268	-0.321	0.385	0.280	0.065	1.000				
SO_4^{2-}	0.034	0.544	0.534	0.652	-0.059	-0.027	0.363	0.472	0.636	0.134	1.000			
Na^+	0.314	0.728	0.748	0.357	0.429	0.402	0.749	0.402	0.164	0.621	0.196	1.000		
F^-	0.006	-0.085	-0.096	0.155	-0.148	-0.012	0.342	0.053	0.226	0.231	0.105	0.295	1.000	

NO ₃ ⁻	-0.031	-0.366	-0.412	-0.324	-0.218	-0.135	0.648	0.417	0.081	0.434	0.112	0.478	0.451	1
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Table 10. Pearson correlation matrix of groundwater parameters for POM season

	pH	EC	TDS	TH	CO ₃ ²⁻	HCO ₃ ⁻	Cl ⁻	Ca ²⁺	Mg ²⁺	K ⁺	SO ₄ ²⁻	Na ⁺	F ⁻	NO ₃ ⁻
pH	1													
EC	-0.065	1												
TDS	-0.069	0.993	1											
TH	-0.319	0.830	0.832	1										
CO ₃ ²⁻	0.670	0.176	0.185	0.117	1									
HCO ₃ ⁻	-0.182	0.587	0.603	0.407	-0.158	1								
Cl ⁻	-0.010	0.839	0.820	0.665	0.227	0.177	1							
Ca ²⁺	-0.287	0.831	0.833	0.924	-0.086	0.439	0.694	1						
Mg ²⁺	-0.242	0.477	0.477	0.714	-0.125	0.174	0.331	0.393	1					
K ⁺	-0.154	0.393	0.402	0.140	-0.391	0.080	0.444	-0.244	0.110	1				
SO ₄ ²⁻	-0.194	0.515	0.527	0.701	-0.151	0.030	0.370	0.554	0.672	-0.048	1			
Na ⁺	0.138	0.640	0.653	0.214	0.414	0.379	0.658	0.332	-0.095	-0.680	-0.043	1		
F ⁻	-0.124	0.042	0.049	0.055	-0.172	0.094	0.658	-0.066	0.254	0.137	0.165	-0.266	1	
NO ₃ ⁻	-0.134	-0.139	-0.103	0.111	0.084	0.658	-0.262	0.232	0.204	-0.010	-0.231	0.276	1	1

CONCLUSIONS

This study assessed groundwater quality in the Mogamureru River Basin by analyzing 46 samples for key physicochemical parameters, including pH, EC, TDS, TH, Cl⁻, HCO₃⁻, Na⁺, Ca²⁺, Mg²⁺, NO₃⁻, F⁻, and SO₄²⁻. Several sampling locations exhibited concentrations exceeding the permissible limits prescribed by WHO (2012) and BIS (2012) drinking water standards. Based on the WQI, 58.7% of samples in the pre-monsoon and 67.4% in the post-monsoon season were classified as unsuitable for drinking and domestic purposes, primarily due to high hardness and intensive agricultural activities. Health risk assessment identified nitrate (NO₃⁻) and fluoride (F⁻) as the major non-carcinogenic pollutants. Although carcinogenic risks for adults, children, and infants remained within acceptable limits, ingestion risks for children and infants exceeded safe thresholds in both seasons. The PIG indicated moderate contamination across most of the study area. PCA revealed that elevated fluoride concentrations primarily originate from geogenic sources such as water–rock interactions involving carbonate rocks and mineral dissolution, along with anthropogenic inputs from fertilizer usage and alkaline groundwater conditions. Elevated nitrate

levels were largely attributed to agricultural runoff, irrigation return flows, and excessive fertilizer application. These findings highlight the dual geogenic and anthropogenic origins of fluoride and nitrate contamination, emphasizing the urgent need for improved sanitation, sustainable agricultural practices, and preventive interventions. Suggested remediation methods include coagulation/precipitation and electrocoagulation, alongside strengthened sanitation infrastructure, responsible farming, and ongoing groundwater monitoring to protect both water quality and public health in the region.

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Author Credit Statement

CK: Manuscript preparation, methodology, sample collection and analysis, remote sensing and GIS mapping. SSG: Manuscript preparation, methodology and revisions. PRK: Methodology, sample collection and analysis. PKB: Revision and manuscript corrections.

Data Availability

Data will be made available on reasonable request from the corresponding author.

Compliance with Ethical Standards

The authors declare that they have no conflict of interest and adhere to copyright norms.

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